

SETU

सेतु — *Sanskrit for bridge.*

The project, the script,
and every question they will ask.

Three parts. What SETU is and what is actually built. Every word you say across seven minutes. And forty-four prepared answers for the eight minutes of questions that follow.



Part 1 is what every team member must know cold. Part 2 is what the two speakers rehearse.
Part 3 is what decides the finale.

The project

PART 1

Everything below is drawn from your repository, not from a slide. Every member of the team should be able to say any of it without looking it up — a jury will address questions to whoever is nearest.

WHAT SETU IS, IN ONE SENTENCE

A cognitive accessibility layer that sits between a page that was never designed for you and the thing you were actually trying to do — for people with dyslexia, dysgraphia, dyscalculia, ADHD and memory-related barriers.

The category line: the market built tools that *read the page*. SETU is the first that *finishes the task*.

THE FOUR SURFACES

Surface	What it is	What it does
SETU Engine	Node · Express · MongoDB	The shared AI orchestration and persistence layer every surface calls. 53 endpoints, 9 cognitive modes, 11 Indian languages. MongoDB is optional — without it everything falls back to local storage and keeps working.
SETU Lens	Chrome extension Manifest V3, v3.2.0	Reshapes the page you are on. 14 content scripts, no build step, no dependencies. Ten reading tools run with no server at all; the AI features add an agent, task chunking and visual mapping.
SETU Sanctuary	React 18 + Vite live on Vercel	The deep-work workspace. Turns a topic, PDF, URL or transcript into an interactive talking mind map. Nine cognitive modes, a reward system, focus sessions, command palette, three-step onboarding.
SETU Go	React Native + Expo Android	Cross-device continuity. SVG touch mind maps, camera OCR document ingestion, cognitive modes, ADHD focus timers, offline-first local storage.

WHAT IS ACTUALLY BUILT

Runs with no server at all — ten tools

Tool	Shortcut	What it does
Bio-Reading	Alt+B	Weights the fixation point of each word to anchor the eye.
Focus Mode	Alt+F	Strips ads, banners, autoplay and chaotic layout into a clean reader view.
Line Focus	Alt+L	Dims everything except the line being read.
Reading Ruler	Alt+H	A tracking guide that holds your place down the page.
Read Aloud	Alt+T	Browser speech synthesis — works with zero engine connection.
Auto Scroll	Alt+S	Paced scrolling at a chosen speed.
Gaze Scroll	Alt+E	Hands-free scrolling from head position. Video never leaves the machine.
Three-step path	Alt+3	Breaks the page into three steps, derived from the page's own fields.

Tool	Shortcut	What it does
Visual map	Alt+M	Maps a chart, table or section from its headings, lists and rows.
Reading themes	popup	Dyslexia-friendly palettes, spacing and line height.

Better with the engine

- **SETU Commander** — reads the page's own reported controls, plans the steps, fills a form, and stops before submitting.
- **The Breathe Protocol** — watches for erratic cursor reversals, scroll thrash and rage-clicking. On detection it dims the page, offers a box-breathing reset, then asks whether to simplify. Deliberately conservative: signals decay, it arms only after a settling period, and it will not re-trigger for several minutes once dismissed.
- **Nine cognitive modes** — Start, Simplify, Learn, Meet, Practice, Write, Guide, Numbers, Listen.
- **Talking mind maps** — branches speak on hover *and* on keyboard focus; a picture mode strips the prose entirely.
- **Natural voice** — Sarvam Bulbul, pipelined by sentence so the first sentence starts in about a second. Falls back to the browser voice automatically, even mid-passage.
- **Parking lot** — offloads an interrupting thought from anywhere without losing your place.
- **Reward engine** — points, levels, streaks and eleven milestones, quiet by default and switchable off without losing progress.

THE NUMBERS TO KNOW COLD

53 API endpoints on the engine	11 Indian languages, end to end
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Figure	Where it comes from
53 endpoints	backend/routes/apiRoutes.js — countable
9 cognitive modes	start, simplify, learn, meet, practice, write, guide, numbers, listen
11 Indian languages	English, Hindi, Bengali, Gujarati, Kannada, Malayalam, Marathi, Odia, Punjabi, Tamil, Telugu — backend/config/languages.js
14 content scripts	chrome-extension/manifest.json, Manifest V3, v3.2.0, store-ready
32-point smoke test	backend/scripts/smoke-test.js — health, model round-trips, persistence, all nine modes, OCR, streaming, voice across all eleven languages
7 extension test suites	imports, DOM, icons, CSS, core runtime, gaze control law, live engine safety
≈30,000 lines · 160 files	Across engine, extension, web workspace and mobile
₹0 infrastructure to date	Render, Vercel, MongoDB Atlas and OpenRouter free tiers, with a deterministic offline engine underneath

THE STACK — SAY EXACTLY THIS

Layer	What it actually is
Engine	Node, Express, MongoDB. Identity is an anonymous per-device token in an x-user-id header.

Layer	What it actually is
Model chain	OpenRouter first (starting on free tiers), then Gemini direct, then OpenAI, then a deterministic offline rule engine. Every AI feature renders something usable off the page in under a second, then upgrades in place.
Extension	Manifest V3, 14 plain JavaScript content scripts operating on the rendered DOM. No build step, no bundler, no dependencies.
Web workspace	React 18 with Vite. Deployed on Vercel.
Mobile	React Native with Expo, TypeScript, Android.
Speech	Sarvam — Bulbul for synthesis, Saaras for transcription. Browser speech synthesis as a zero-cost fallback.

Never inflate this. There is no WebAssembly, no Rust, no Supabase, no pgvector and no client-side AES-256 in the repository. If a judge asks and you name one of those, every other claim you made becomes unverified. The real stack is impressive and does not need help.

THE CLINICAL REVIEW — 17 AUGUST 2026

You took SETU to **Dr. Prakhar Jain**. He contradicted the assumption the product was built on, and the web application was reworked against his findings. This is the single highest-scoring thing you have.

What he established	What you changed
"Dyslexia" is three separate conditions — dyslexia (reading), dyscalculia (numbers), dysgraphia (writing). A person may have one, two or all three.	Onboarding lists all three separately. A dedicated Numbers mode was added.
Bigger fonts and reading aids are largely ineffective — the difficulty is comprehension and decoding, not eyesight.	Bio-Reading demoted from headline feature to an honestly labelled comfort option. A new Comprehension Support section leads with audio and diagrams.
Mind maps only help if they carry no text, or if the text is spoken when the cursor reaches it.	Branches speak on hover <i>and</i> on keyboard focus. Picture mode strips the prose and moves the detail to the voice.
Maths must be taught with countable physical objects inside a story — apples and bananas, the way special education does it.	Numbers mode renders the actual objects on screen, one step at a time, narrated aloud.
ADHD needs work made interesting and gamified — milestones and points, not long text reports.	A full reward engine: points, levels, streaks and eleven milestones, summarised as one ring.
Working memory is impaired; long-term memory is not.	The parking lot (Alt+P) offloads an interrupting thought from anywhere without losing your place.
Around 40% also experience anxiety or depression; add listener support so they can express frustration.	Listen — a reflective listener with a hardcoded crisis path and real helplines.

And the part that matters commercially: Dr. Jain has since offered to build a version of SETU for his hospitals. That is inbound demand from a clinician, before launch. It belongs on slide 5 and in the close.

SAFETY AND PRIVACY — THE THREE THINGS THAT ARE TRUE

- **The agent never submits.** Three rules asserted in the test suite: no step may target a control the page did not report, every submit step carries a confirmation requirement, and no personal data is ever invented to fill a field.

- **Crisis language never reaches a model.** Detected server-side before any model call, answered from a fixed reviewed script carrying Tele-MANAS 14416, KIRAN, AASRA and an international directory. Never sampled, never scores points, deliberately not translated.
- **Almost nothing is stored.** No name, no email, no account. Overwhelm signals are computed in browser memory from cursor and scroll movement — no camera, no biometrics, no keystroke logging. Journal entries are written to a local-only key and never mirrored to the database. API keys live server-side only.

THE BUSINESS, IN ONE PAGE

Question	Answer
What does one user cost?	₹25 a month at paid tiers — ₹8 AI reasoning, ₹15 natural voice, ₹2 storage, ₹0 for the ten local tools. Today it is ₹0 , because the model chain starts on free tiers.
Who pays?	Institutions, not students. Campus at ₹199 per student per year , licensed campus-wide. Employers at ₹499 per employee per year from existing accessibility and DEI budgets. Individuals free forever, ₹299 a month premium.
Why campus-wide?	Because the university cannot identify who needs it. India registers 0.1% of students as disabled against a far higher true prevalence — a per-seat licence only reaches the students already being supported.
What is the margin?	A 5,000-student campus pays ₹9.95 lakh and costs about ₹2.2 lakh to serve at 12% activation. 78% gross margin.
When do you break even?	Year-one build and run is ₹18 lakh. Two campus licences cover it.
How big does it get?	Year 1: 10 campuses, ₹1.0 crore. Year 2: 100 campuses, ₹10 crore. Year 3: 500 campuses, ₹49.8 crore — which is 1% of India's 48,246 colleges.

WHAT IS NOT BUILT — SAY IT BEFORE THEY FIND IT

- **No institutional dashboard.** A university will want one by month two of a pilot.
- **The app's own menus are English-only.** SETU writes and speaks in eleven languages; its own chrome does not yet.
- **No outcome data.** You have a working product and an established intervention — not your own measured results. That is exactly what the pilot is for.

The presentation

PART 2

Seven minutes, hard stop. Written at roughly 145 words per minute — deliberately unhurried, because the subject is overload and your delivery is part of the argument. Read it aloud against a stopwatch four times before you change a word.

THE CLOCK

Slide	Clock	Runs	Speaker	The one job this slide has
1	0:00–0:40	0:40	Speaker 1	Make them feel it. Name the product. Plant the reframe.
2	0:40–1:40	1:00	Speaker 1	Size the problem. Kill the three existing categories.
3	1:40–2:45	1:05	Speaker 1	The architecture, and the clinical rewrite.
4	2:45–5:15	2:30	Speaker 2	Prove it works. Five beats.
5	5:15–6:15	1:00	Speaker 2	Prove it survives outside a hackathon.
6	6:15–6:55	0:40	Speaker 1	The reframe returns. The ask. Then silence.

Finishes at 6:55. The one-minute warning at 6:00 lands mid-way through slide 5 — do not react to it and do not speed up. If you are behind at 5:15, cut demo beat 5, never the business slide and never the ask.

SLIDE 1 • 0:00 – 0:40 • SPEAKER 1

“Good morning respected judges and fellow innovators.

Imagine opening a website and trying to read an article... but the letters start moving, lines get mixed up, and your brain gets overwhelmed within seconds.

For millions of people with ADHD and dyslexia, this is not imagination — it is their daily reality.

Today we present SETU — **a solution that fixes the web, not the person.**

Setu is Sanskrit for *bridge*. It is a cognitive accessibility layer that sits between a page that was never designed for you, and the one thing you were actually trying to do.”

ACTION. Deliver the first three lines slowly and evenly — this is the only moment in seven minutes where you are allowed to be quiet and warm. Take a full second of silence after “*daily reality*”. Make eye contact with the most senior person in the room on “*fixes the web, not the person*” and hold it for one beat. Do not smile through the hook; smile after it.

“There is a name for the moment I just described. Cognitive scientists call it the **Wall of Awful** — the instant executive function stops, because the interface is asking for more than working memory can hold. It is not a failure of intelligence. It is a failure of design.

Dyslexia affects around 780 million people worldwide. One in four people with ADHD have it too. Across the population, fifteen to twenty per cent of us are neurodivergent in some way.

In Indian higher education alone, that is about **nine million students**. One in five.

Now the number the system can actually see. **Fifty-four thousand and eighty**. That is every student in the country formally registered with a benchmark disability — all disabilities, not just these. Nought point one per cent.

[pause — two full seconds]

So nearly nine million students are getting through this alone. No assessment, no accommodation, and no idea that the problem was never them.

And the web is moving away from them, not towards them. WebAIM tested a million home pages this year: **95.9 per cent fail** accessibility standards. That is worse than last year — the first reversal in seven years.

Why has nothing solved it? Screen readers read the page aloud — the overwhelm is identical, just louder. Reading tools restyle the text — the task is exactly as hard. And AI chatbots answer questions *about* the page, then forget you the moment the tab closes.

The market built tools that read the page. Nobody built an engine that finishes the task.”

ACTION. Speed up very slightly through the statistics — you are simulating the pile-up. Then stop dead for two seconds after “nought point one per cent” and let the sliver on the slide do the work. On the last line use your hands: left hand low on “*read the page*”, right hand raised and open on “*finishes the task*”.

IF BEHIND. Cut the WebAIM sentences. Keep the two numbers, the pause, and the last line.

“Before I show you what we built — one pattern worth naming.

Kerb ramps were cut into pavements for wheelchair users. Today the people who use them most are parents with prams and travellers with suitcases. Text messaging was built for deaf users. Captions were built for the hard of hearing, and are now how most people watch video on a train. **Every one of those started as an accommodation and ended as a market.** We are building the curb cut for attention.

SETU is one engine behind three surfaces. A Chrome extension that reshapes the page you are on. A web workspace that turns a document into something you can actually hold in your head. And an Android app, so the task survives leaving your desk. Fifty-three endpoints, nine cognitive modes, eleven Indian languages — deployed and running right now.

Three things make it hard to copy.

One — it is local-first. Ten of the reading tools never touch a server. They cost us nothing per use, and they keep working with the network off.

Two — it degrades instead of failing. Every AI feature draws a usable answer off the page in under a second, then upgrades it when the model replies. A tool that shows a spinner and then an error is a tool nobody opens twice.

Three — and this is the one I would ask you to remember. On the seventeenth of August we took SETU to Dr. Prakhar Jain. He told us that bigger fonts and reading aids do not solve dyslexia, because the difficulty is comprehension, not eyesight. He was contradicting our headline feature.

So we demoted it. And we rebuilt: a Numbers mode for dyscalculia that uses countable objects inside a story, a picture mode that carries no text at all, and a crisis path that never reaches a model.

We did not add a clinician to a slide. We let one change the product.”

ACTION. The curb-cut section should sound like an aside — slower, conversational, as though letting them in on something. Then snap back to pace for the architecture. On “we demoted it,” slow right down. This is the highest-scoring sentence in your seven minutes and most teams would never say it out loud. Speaker 2 walks to the laptop during the final line — do not narrate the handover.

IF BEHIND. Cut the curb-cut paragraph and save it for Q&A. Never cut the clinical rewrite.

“This is Meera. Second year, and it is the last day to renew her scholarship. Let us do it with her.”

BEAT 1 • The portal (0:00 – 0:30)

“Fourteen fields. Three warnings. No indication of what to do first. For most of us this is annoying. For Meera this is where the task ends — and the deadline is today.”

BEAT 2 • Alt + 3 (0:30 – 1:05)

“One keystroke. Three steps, in under a second — and notice what did *not* happen. Nothing left this browser. No server, no API key, no cost to us at all. This is one of the ten local tools, and it is why our margin story works.”

FALLBACK. If the chunker stalls, press **Alt+F** for Focus Mode — fully local, makes the identical point. Say: *“Focus Mode, same idea — the page gets quieter before anything gets smarter.”*

BEAT 3 • Commander (1:05 – 1:45)

“Now the agent. I ask it to fill the form. It reads the page’s own reported controls, plans the steps, fills them — and then it **stops**. It will not submit. Anything irreversible needs a human to say yes.

That is not a policy we wrote in a slide. It is asserted in our test suite: no step may target a control the page did not report, every submit step carries a confirmation requirement, and no personal data is ever invented to fill a field.”

ACTION. This is the trust beat and the answer to the question every technical judge is already forming. Slow down. Let the cursor sit on the unsubmitted form for two full seconds before speaking again. Silence over a stopped agent is more convincing than any sentence.

BEAT 4 • The voice (1:45 – 2:15)

“Dr. Jain told us mind maps only help if they carry no text — or if the text is spoken the moment the cursor reaches it. So: hover a branch, and it speaks. In Tamil. Eleven Indian languages, and the data schema underneath stays in English, so nothing silently breaks when the language changes.”

FALLBACK. If the voice is slow, the browser engine takes over automatically. Say it out loud: *“— and there is the fallback. The user never hears silence.”* The failure demonstrates the feature.

BEAT 5 • The handoff (2:15 – 2:30)

“Same student, same task, on the phone. It picks up mid-sentence. Working memory is the thing that is actually impaired here — so SETU carries the context, and Meera does not have to.”

CLOSE ON HER NAME. *“Meera submits the form. On her own. That is the entire product.”* Then return to the deck.

“Now the part that decides whether this survives past this room.

One active user costs us **twenty-five rupees a month**. Eight for AI reasoning, fifteen for the natural voice, two for storage — and nothing at all for the ten local tools, because they never call us.

Today it costs us **zero**. The model chain starts on free tiers and ends in a deterministic offline engine, so our infrastructure spend to date is nil. Four production surfaces, built by five final-year students, for the cost of our time.

We sell to campuses at **₹199 per student per year, licensed campus-wide** — because the university cannot possibly know which students need it. Ninety-eight per cent of them were never assessed. Campus-wide is not a pricing trick; it is the only honest way to price an invisible population.

Year one costs us eighteen lakh to build and run. **Two campuses cover it**.

And this is not theoretical. Dr. Prakhar Jain — the specialist who reviewed us — has asked us to build a version of SETU for his hospitals. That came to us. Before we launched.”

ACTION. The most number-dense minute in the pitch, so slow down rather than speeding up. Land **₹25, ₹0** and **two campuses** as three separate beats with a half-second of air around each. The warning bell falls somewhere in here — ignore it completely.

“Every one of those nine million students has been told, in one way or another, that they are the problem.

They are not. **The interface was the disability. Not the person.**

Your digital inclusion programme has reached 8.4 million people and is aiming at ten million by 2030. Neurodivergent learners are not named in it yet. We would like to be how you reach them.

Three asks. **One pilot campus**, through a Capgemini client, this semester. **One introduction** to a university disability support office. And **back SETU in this finale** — we will publish the numbers either way.

Thank you.”

ACTION. Land “*The interface was the disability. Not the person.*” then stop for a full second. Say the three asks on your fingers — one, two, three — because a jury that can count your asks can grant one. Then say thank you, stand still, and **say nothing else**. Do not fill the silence. The last sentence they hear is the one that follows them into scoring.

The questions

PART 3

Forty-four questions across eight categories. Eight minutes of Q&A is longer than your presentation and it is where the finale is decided. Answer in three sentences and then stop — a short confident answer scores higher than a complete one.

A • IMPACT AND EVIDENCE

A1. What evidence do you have that this actually helps?

Two kinds. Structured task breakdown and single-step presentation are established practice in executive-function coaching and special education — we did not invent the intervention, we automated it. And we ran the product past a clinician on the seventeenth of August, which changed the build substantially. What we do not yet have is outcome data, and that is exactly what the pilot is for.

Established intervention, automated. Outcome data is what the pilot buys.

A2. What did the clinical review actually change?

More than we expected. Dyslexia was separated into three conditions in onboarding, with a new Numbers mode for dyscalculia that renders countable objects inside a story. Bio-Reading was demoted from headline feature to an honestly labelled comfort option. Mind map branches now speak on hover and on keyboard focus, and a picture mode strips the prose entirely. A reward system replaced text reports for ADHD, a parking lot was added for impaired working memory, and a reflective listener was added because roughly forty per cent of this population also experiences anxiety or depression.

He contradicted our flagship. We rebuilt around him.

A3. How will you measure whether it worked?

Three numbers, all measurable inside one semester: time to first action, task completion rate on dense multi-step forms, and the proportion of tasks completed without asking another person for help. The third is the real outcome — independence, not usage.

Time to first action, completion rate, independent completion.

A4. Isn't this a niche market?

Kerb ramps were built for a few hundred wheelchair users and are now used by everyone with a suitcase. Text messaging was built for deaf users. Captions were built for the hard of hearing and are how most people watch video in public. Accessibility features have a consistent history of outgrowing the population they were built for — and in the meantime, one in five students is not niche.

The curb-cut effect. Accommodations become markets.

A5. Are you diagnosing people?

No, and we are deliberate about it. Onboarding asks what tends to get in your way, not what you have been diagnosed with, and the result is a preference profile rather than a medical record. Nothing in the product asserts a condition about anyone.

A preference profile, not a diagnosis.

A6. Does this replace human disability support staff?

It extends their reach rather than replacing them. A support office typically has a handful of staff for thousands of students, so they triage. SETU handles the moment-to-moment friction of everyday digital tasks so human time goes to the higher-level work only a person can do.

It extends the support office, it does not replace it.

B • BUSINESS MODEL AND BUYER

B1. Who actually pays?

Institutions, not students. A university licenses campus-wide at ₹199 per student per year; an employer buys per seat at ₹499 from an accessibility or DEI budget that already exists. Individuals get the local tools free forever, because those genuinely cost us nothing to serve.

Universities and employers, from budgets that already exist.

B2. Why campus-wide rather than per seat?

Because the buyer cannot identify the users. India registers 0.1% of its students as disabled against a far higher true prevalence, so a per-seat licence would only cover students already receiving support — the group that needs us least. Campus-wide is the only structure that reaches the rest, and it works commercially because most licensed students only ever use the tools that cost us nothing.

You cannot buy seats for people nobody has identified.

B3. How did you arrive at ₹199?

Bottom-up from cost. An active user costs ₹25 a month, activation in comparable campus tools runs around twelve per cent, so a five-thousand-student campus costs roughly ₹2.2 lakh a year to serve and prices at ₹9.95 lakh. That is a 78% gross margin, and the headroom funds the permanently free individual tier.

Cost-up, not market-down. 78% gross margin.

B4. Why is retail ₹299 a month when institutions pay ₹199 a year?

Because institutions buy five thousand seats of which about twelve per cent activate, so the effective price per active student is around ₹1,650 a year — roughly ₹138 a month. Against ₹299 retail that is a 2.2× volume discount, which is ordinary enterprise pricing.

Per active user it is a 2.2× discount, not 18×.

B5. Institutional sales cycles are long. How do you get in?

We do not start at central procurement. We start at the disability support office, which has a statutory mandate under the RPwD Act, a budget line, and more students than staff. A semester pilot is a far shorter decision than an IT purchase, and the pilot's outcome numbers unlock the second campus.

Disability support office first, procurement second.

B6. Realistically, what does year one look like?

Ten campuses and about one crore, with break-even at two. We would rather present a number we can hit in two semesters than a curve we cannot defend.

Ten campuses, ₹1 crore, break-even at two.

B7. What is the enterprise angle for a company like Capgemini?

Major employers have largely solved how to hire neurodivergent talent — JPMorgan's programme now spans more than forty roles across nine countries, and SAP, HPE and Microsoft have all reformed hiring to reach this group. What none of them has solved is making the internal tooling usable once those people arrive. That is a retention problem with a software answer, sitting inside an existing DEI budget.

They solved hiring. Nobody solved the tooling.

C • COST, MARGIN AND SCALE

C1. What does this cost at a million users?

Most of a million users cost nothing, because the ten local tools never reach our servers. At twelve per cent monthly-active on AI features, a million licensed users is roughly ₹3.6 crore a year in inference and voice, against ₹19.9 crore of revenue at campus pricing.

The architecture is why the margin holds at scale.

C2. You are running on free model tiers. That is not a business.

Agreed, and we do not present it as one. We present it as evidence that the system degrades: the free tier is the top of a chain that ends in a deterministic offline engine. So we already know precisely what happens when it disappears — the ₹25 figure on our slide is that scenario, fully costed.

The free tier is a fallback chain, not a business model.

C3. Voice is your largest cost line. What if Sarvam raises prices?

Three answers. Clips are cached at both ends, so a repeated label is synthesised once. The browser's own voice is a working fallback at zero cost and the engine switches to it automatically, even mid-passage. And text-to-speech is the most competitive segment in AI right now, so the price risk points down.

Cached, fallback-covered, and in a deflating market.

C4. What has this cost you to build so far?

Zero in infrastructure. Render, Vercel, MongoDB Atlas and OpenRouter free tiers, with a deterministic offline engine for when they rate-limit. Five students built four production surfaces for the cost of their time.

₹0 of infrastructure. Four surfaces. Five people.

C5. What if activation is much higher than twelve per cent?

Then margin compresses and the price moves — at twenty per cent activation, campus pricing goes to about ₹299 and the model still works. Activation is our least certain assumption and we would rather find out on one campus than on five hundred.

Our softest number, and the pilot is designed to test it.

C6. A 78% gross margin sounds too good. Justify it.

It is a normal software margin, and it is high specifically because the architecture keeps most users off our servers. The eighty-eight per cent of licensed students who only use the local reading tools cost us literally nothing. Had we built this cloud-first, the same product would carry a fraction of that margin.

The margin is a consequence of the architecture.

D • TECHNOLOGY AND ARCHITECTURE

D1. What is your stack, precisely?

Node and Express with MongoDB for the engine — fifty-three endpoints, nine cognitive modes. A Manifest V3 Chrome extension with fourteen content scripts and no build step. React with Vite for the web workspace. React Native with Expo for Android. Sarvam for speech in eleven Indian languages, and a model chain that starts on OpenRouter and falls through to Gemini and then to a deterministic offline engine.

Know this cold. Never inflate it.

D2. How do the local tools work with no server at all?

They are content scripts operating directly on the rendered DOM in the browser's own JavaScript engine — bio-reading weighting, focus mode, line focus, the reading ruler, auto-scroll, themes, and read-aloud through the browser's speech synthesis. No network call is involved in any of them, which is why they work offline and cost us nothing per use.

Content scripts on the rendered DOM. No network, no cost.

D3. Does it work on JavaScript-heavy single-page applications?

Yes, because we read the rendered DOM as the browser currently has it, not the source, so dynamically loaded content behaves exactly like static content. The extension runs on every http and https page with no per-site configuration.

We read the rendered DOM, so SPAs are not a special case.

D4. What is the latency?

Local tools are instant. The three-step path and the visual map render from the page in under a second and then upgrade in place when the model replies, typically two to four seconds later. Read-aloud is pipelined by sentence, so the first sentence starts in about a second instead of after the whole passage has synthesised.

Under a second locally, then it upgrades in place.

D5. What was your hardest technical problem?

Making AI features usable on rate-limited free models. A tool that shows a spinner and then an error is a tool nobody opens twice — so every AI feature now renders something real off the page in under a second and upgrades when the model replies. That constraint made the product better than an unlimited budget would have.

The constraint produced the render-then-upgrade pattern.

D6. How do you know it works? What is your test coverage?

A thirty-two point automated smoke test on the backend covering health probes, model round-trips, persistence, all nine modes, document OCR, streaming, and voice synthesis across all eleven languages — including an assertion that Hindi and Tamil responses keep their schema enums in English, because a model that helpfully translates a priority label silently breaks the UI. The extension has seven separate suites, none of which needs a bundler.

32 backend checks, 7 extension suites, and we test the failure modes.

D7. How do you handle eleven languages without the whole thing breaking?

One configuration file is the single source of truth for both halves — the speech language code and the instruction that makes the model answer in that language. Two details matter in practice: schema enum values stay in English so the client can compare them, and the sentence chunker knows the danda, because without it a Hindi paragraph reads as one enormous sentence and gets cut mid-clause.

One source of truth, English enums, and the chunker knows the danda.

E • AI SAFETY AND RELIABILITY

E1. Autonomous web agents hallucinate and break on real forms. How is yours different?

Three hard rules, all asserted in the test suite rather than promised in a slide: no step may target a control the page did not report, every submit step carries a confirmation requirement, and no personal data is ever invented to fill a field. The agent reads the page's semantic accessibility surface rather than guessing from pixels — the same approach Microsoft ships in Playwright MCP. It is an assisted execution engine, not an autonomous one.

Assisted execution, human-in-the-loop, asserted in tests.

E2. What if the page changes while the agent is mid-task?

The plan is validated against the controls the page currently reports, so a control that has vanished fails that step with a specific error code the UI keys its guidance off — it does not click something else instead. We would rather stop and tell the user than act on a stale plan.

It fails loudly rather than acting on stale state.

E3. How do you stop the AI getting the content wrong?

We keep its job narrow and structural. It restructures content that is already on the page into a fixed JSON schema — it is never asked to generate new facts. Because everything renders from a schema, a malformed answer fails visibly instead of quietly producing something plausible and wrong.

Restructuring, not generating. Schema failures are visible.

E4. What about false positives — you interrupt someone who was working fine?

The detector is deliberately conservative, because a false positive interrupts someone who was fine and that is worse than a missed detection. Signals decay continuously, the detector only arms after a settling period, and once dismissed it will not re-trigger for several minutes. The prompt itself is a dismissible overlay, not a modal.

Conservative by design; a false positive is the worse error.

E5. Someone in crisis uses your reflective listener. What happens?

Crisis language is detected server-side before any model call, and the reply comes from a fixed, reviewed script carrying Tele-MANAS on 14416, KIRAN, AASRA and an international directory. It is never sampled from a model, never depends on the AI being reachable, and never scores points in the reward system. It is also deliberately not translated — a subtle translation error is unacceptable in that specific message, so it stays in English and leads with phone numbers, which need no translation.

Detected before the model, answered from a fixed script, never sampled.

F • PRIVACY, ETHICS AND COMPLIANCE

F1. You are reading behaviour and documents. What do you store, and where?

Identity is an anonymous per-device token — no name, no email, no account required. The overwhelm signals are computed in browser memory from cursor and scroll movement and are never transmitted; there is no camera, no biometrics and no keystroke logging. Journal entries in the reflective listener are written to a local-only key and are never mirrored to the database, and the API keys live server-side only so the browser never sees them.

Anonymous by default. The most sensitive data never leaves the device.

F2. How does that stand up to GDPR or an institutional DPA?

Our strongest position is data minimisation — we hold almost nothing personally identifying, which removes most of the surface an assessment would examine. For an institutional deployment we would expect to sign a data processing agreement, complete a VPAT, and add tenant-level deletion and export. Those are pilot deliverables and we would rather scope them with a real customer than guess at them now.

Minimisation first; DPA and VPAT are pilot deliverables.

F3. What accessibility standards do you meet?

WCAG 2.1 AA for structure and contrast, and Universal Design for Learning for presenting the same content in the form a given person processes best. The mobile client was checked against AA contrast, font scaling, screen-reader labels and forty-eight pixel touch targets. We have not yet had an independent audit, which is budgeted in year one.

WCAG 2.1 AA and UDL. Independent audit budgeted, not done.

F4. Is the AI trained on user data?

No. We call third-party models through their APIs and route around any tier that reserves the right to train on payloads. We hold no training pipeline of our own, and given who our users are, we do not intend to build one.

No training on user data, and no ambition to.

F5. What is the risk of a tool like this being paternalistic?

It is a real risk and it drove several design decisions. Every intervention is offered and dismissible, never forced. The reward system is quiet by default and can be switched off entirely without losing progress. And onboarding asks about preferences rather than diagnoses, because the moment a tool tells someone what they are, it has stopped being an accommodation.

Offered, never forced. Preferences, never diagnoses.

G • COMPETITION AND MOAT

G1. Microsoft Immersive Reader, Speechify and Texthelp all exist. What stops them crushing you?

They ship reading. Immersive Reader has been free and built into Edge and Office for years, and 95.9% of the web still fails accessibility standards — which tells you reading was never the binding constraint. Nobody has shipped the layer after reading: detecting when executive function stops, restructuring the task, acting on the page with confirmation, and carrying context across devices. And they will not build eleven Indian languages with a dyscalculia mode, because that market is too small for them and exactly right for us.

They solved reading. Reading was not the constraint.

G2. What genuinely stops a well-funded team copying this in six months?

Any one feature is copyable. What compounds is the combination — behavioural detection makes the intervention timely, a persistent cross-surface profile makes each one better than the last, structured task breakdown produces the outcome the institution is buying, and eleven Indian languages make it usable where we win first. The more durable moat is distribution: a support office that has run a semester pilot, has our numbers and has trained its staff does not re-procure casually.

The loop compounds, and the relationship is the real moat.

G3. Why would a university that already owns Read&Write buy you?

Because a university that owns a reading tool still has students who cannot start the form. We are not competing on text-to-speech. Several campuses will run both, and that is fine — we are a different line item solving a different failure.

Different category, different budget line, coexists happily.

G4. Is there anything defensible in the technology itself?

Honestly, the individually defensible piece is smaller than the product. What is hard is the accumulated judgement — the detector being conservative enough not to annoy people, the render-then-upgrade pattern that keeps AI features usable on rate-limited tiers, the schema enums staying in English so eleven languages do not break the UI. None of that is patentable and all of it took a clinician and several rebuilds to arrive at.

Accumulated judgement plus distribution, not a patent.

H • TEAM, ROADMAP AND THE HARD ONES

H1. What is not built yet?

Three things, and we would rather say them than have you find them. There is no institutional dashboard, which a university will want by month two of a pilot. The app's own menus are still English-only — we change what SETU writes and says in eleven languages, not its own chrome. And we have no outcome data, only a working product.

No dashboard, English-only UI chrome, no outcome data yet.

H2. If you win today, what happens in ninety days?

A 250-student cohort with one university's disability support office, measuring the three metrics on our closing slide. The institutional dashboard, because it is what stands between a pilot and a renewal. And an internal trial with a Capgemini employee resource group if you will have us — a workplace cohort tests a different failure mode than a campus one.

250-student pilot, the dashboard, and an ERG trial.

H3. You all graduate this year. Why should we believe this continues?

It is a fair question and the honest answer is that a pilot changes the decision. Right now this is five final-year students with a running product and no obligation to anyone. One campus with a signed semester pilot, and a clinician already asking for a hospital build, is a materially different choice — which is exactly why our ask is a pilot rather than a prize.

The pilot converts this from a project into a company.

H4. What is the weakest part of this pitch?

That we are asking you to believe a product works before anyone has measured whether it does. Everything technical we have shown you is running and you can check it. The impact claim rests on an established intervention and one clinician's review — not on our own data, because we do not have any yet. That is the gap, and closing it is the only thing we are asking for.

Answer this one honestly. It is the highest-scoring answer in the set.

THE FIVE YOU MUST NOT IMPROVISE. Rehearse these word for word until they are automatic — they are the ones where a fumble is expensive:

D1 (your stack) · E1 (agent safety) · F1 (what you store) · B2 (why campus-wide) · H4 (your weakest point).

WHEN YOU DO NOT KNOW. Say so, then convert it into a commitment: *"I don't have that number. I'd rather not guess — can I follow up after the session?"* That costs almost nothing. A guess that turns out to be wrong costs you everything you said before it. Agree in advance who answers what — one person on business and cost, one on architecture and safety — and never interrupt each other.